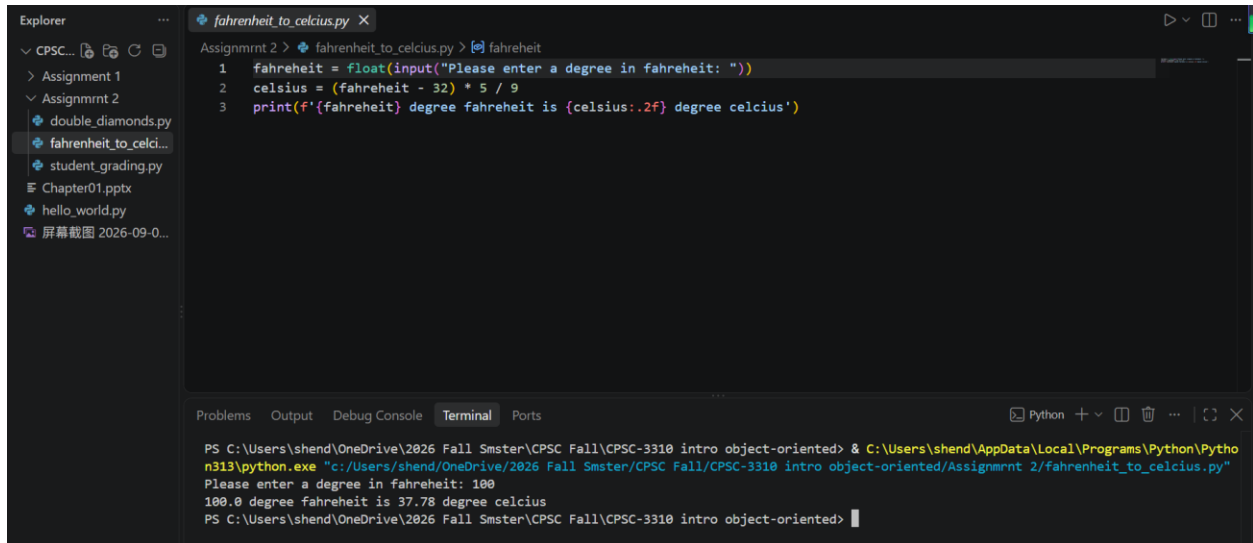


Assignment 2

fahrenheit_to_celcius.py

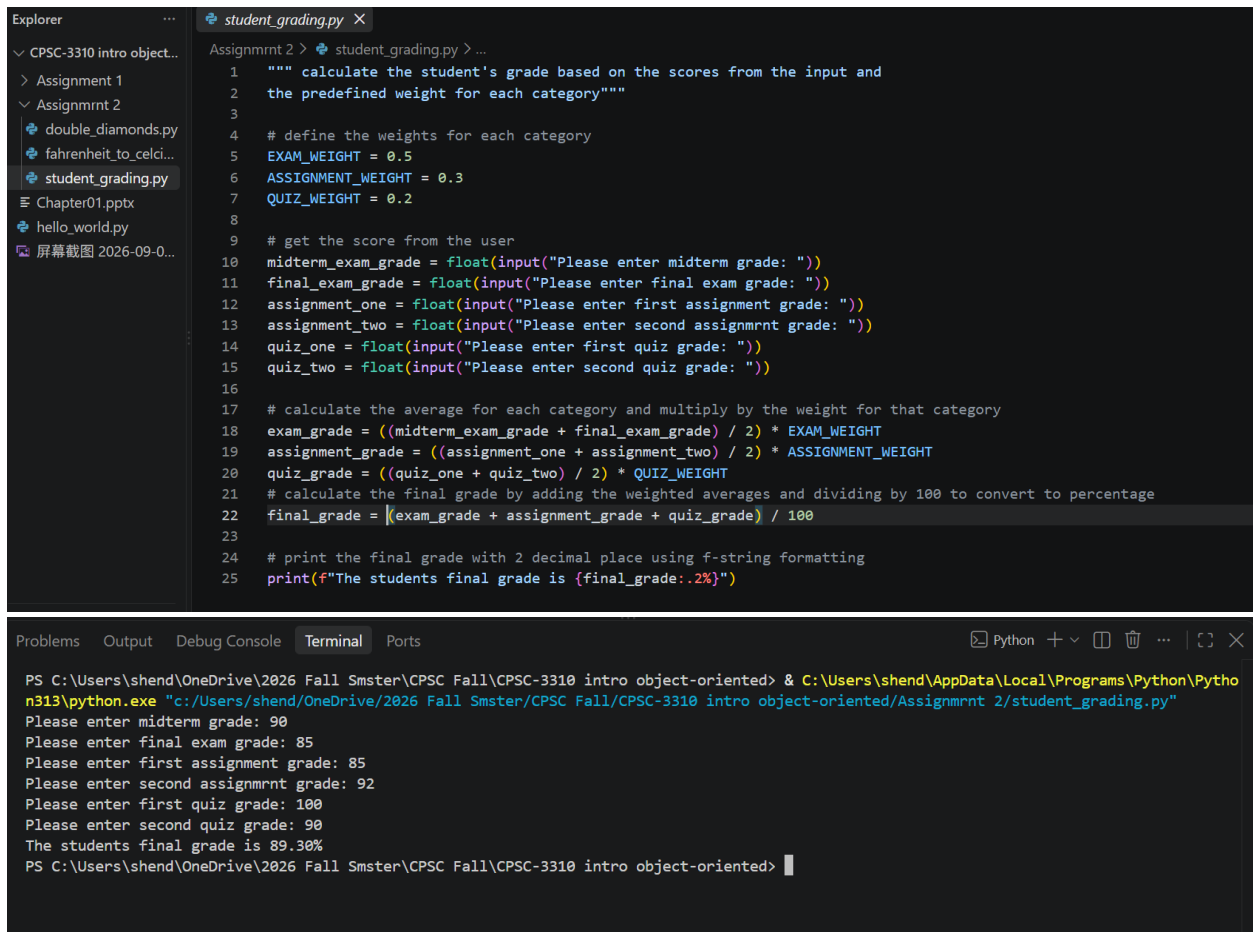


The screenshot shows a VS Code editor with the Explorer sidebar on the left. The file explorer shows a project structure with folders 'CPSC...' and 'Assignment 2'. Inside 'Assignment 2', there are files 'double_diamonds.py', 'fahrenheit_to_celcius.py', 'student_grading.py', and 'Chapter01.pptx'. The 'fahrenheit_to_celcius.py' file is selected and its code is displayed in the main editor. The code is a Python script that takes a Fahrenheit temperature as input and converts it to Celsius. The terminal at the bottom shows the command to run the script and the output, which is '100.0 degree fahrenheit is 37.78 degree celcius'.

```
1 fahrenheit = float(input("Please enter a degree in fahrenheit: "))
2 celsius = (fahrenheit - 32) * 5 / 9
3 print(f'{fahrenheit} degree fahrenheit is {celsius:.2f} degree celcius')
```

```
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> & C:\Users\shend\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 2/fahrenheit_to_celcius.py"
Please enter a degree in fahrenheit: 100
100.0 degree fahrenheit is 37.78 degree celcius
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented>
```

student_grading.py



The screenshot shows a VS Code editor with the Explorer sidebar on the left. The file explorer shows a project structure with folders 'CPSC-3310 intro object...' and 'Assignment 2'. Inside 'Assignment 2', there are files 'double_diamonds.py', 'fahrenheit_to_celcius.py', 'student_grading.py', and 'Chapter01.pptx'. The 'student_grading.py' file is selected and its code is displayed in the main editor. The code is a Python script that calculates a student's final grade based on midterm exam, final exam, assignment, and quiz scores. The terminal at the bottom shows the command to run the script and the output, which is 'The students final grade is 89.30%'.

```
1 """ calculate the student's grade based on the scores from the input and
2 the predefined weight for each category"""
3
4 # define the weights for each category
5 EXAM_WEIGHT = 0.5
6 ASSIGNMENT_WEIGHT = 0.3
7 QUIZ_WEIGHT = 0.2
8
9 # get the score from the user
10 midterm_exam_grade = float(input("Please enter midterm grade: "))
11 final_exam_grade = float(input("Please enter final exam grade: "))
12 assignment_one = float(input("Please enter first assignment grade: "))
13 assignment_two = float(input("Please enter second assignment grade: "))
14 quiz_one = float(input("Please enter first quiz grade: "))
15 quiz_two = float(input("Please enter second quiz grade: "))
16
17 # calculate the average for each category and multiply by the weight for that category
18 exam_grade = ((midterm_exam_grade + final_exam_grade) / 2) * EXAM_WEIGHT
19 assignment_grade = ((assignment_one + assignment_two) / 2) * ASSIGNMENT_WEIGHT
20 quiz_grade = ((quiz_one + quiz_two) / 2) * QUIZ_WEIGHT
21 # calculate the final grade by adding the weighted averages and dividing by 100 to convert to percentage
22 final_grade = (exam_grade + assignment_grade + quiz_grade) / 100
23
24 # print the final grade with 2 decimal place using f-string formatting
25 print(f"The students final grade is {final_grade:.2%}")
```

```
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> & C:\Users\shend\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 2/student_grading.py"
Please enter midterm grade: 90
Please enter final exam grade: 85
Please enter first assignment grade: 85
Please enter second assignment grade: 92
Please enter first quiz grade: 100
Please enter second quiz grade: 90
The students final grade is 89.30%
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented>
```

double_diamonds.py

```
1  """this program is generate two diamonds using turtle graphics"""
2
3  import turtle
4
5  # predfine the diamond side lenght - fill the code
6  # predfine the fill color - fill the code
7
8  # setup the turrtle window
9
10 turtle.setup(500,600)
11
12 # steup the turtle
13 turtle.hideturtle()
14 turtle.fillcolor("blue")
15 turtle.speed(1)
16
17 LEFT_SIDE_LENGTH = 100
18 RIGHT_SIDE_LENGTH = 100
19
20 # draw the lest-side diamond first using the predefined SIDE_LENGTH - fill the code
21 turtle.begin_fill()
22 # fill the code to draw the left-side diamond
23 turtle.left(135)
24 turtle.forward(LEFT_SIDE_LENGTH)
25 turtle.left(90)
26 turtle.forward(LEFT_SIDE_LENGTH)
27 turtle.left(90)
28 turtle.forward(LEFT_SIDE_LENGTH)
29 turtle.left(90)
30 turtle.forward(LEFT_SIDE_LENGTH)
31 turtle.end_fill()
32
33 # draw the right-side diamond next - fill the code
34
35 turtle.begin_fill()
36 turtle.left(0)
37 turtle.forward(RIGHT_SIDE_LENGTH)
38 turtle.right(90)
39 turtle.forward(RIGHT_SIDE_LENGTH)
40 turtle.right(90)
41 turtle.forward(RIGHT_SIDE_LENGTH)
42 turtle.right(90)
43 turtle.forward(RIGHT_SIDE_LENGTH)
44 turtle.end_fill()
45
46 turtle.done()
```

